

Alireza Laalmanesh

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RESEARCH INTERESTS

- Emergency Medical Services optimization, real-time ambulance dispatch, dynamic relocation, and resource allocation under uncertainty.
- Discrete-event simulation, stochastic modeling, online optimization, sequential decision-making, and learning-guided combinatorial optimization.

EDUCATION

University of Waterloo

Master of Mathematics, Computational Mathematics (Co-op)

Waterloo, Canada

Sep 2025 – Present

- Recipient of the **i-Comp-Data Fellowship**; fellowship advisors: Dr. David Del Rey Fernández and Dr. Leia Minaker.
- Research interests: numerical optimization, stochastic modeling, deep learning for urban data, and discrete-event simulation.

Amirkabir University of Technology (Tehran Polytechnic)

B.Sc. in Industrial Engineering

Tehran, Iran

2019 – 2024

- GPA: 3.5/4.0. Thesis: *Design of a Model for the Operating Room Scheduling Problem with Endogenous Uncertainty*.

RESEARCH AND PROFESSIONAL EXPERIENCE

Polytechnique Montréal

Co-op Research Assistant

Montréal, Canada

Jun 2026 – Sep 2026

- Conducted computational research on Emergency Medical Services dispatch and dynamic ambulance operations.
- Designed and executed discrete-event simulation experiments to evaluate reinforcement-learning dispatch policies using operational EMS data.
- Developed research directions related to interpretable ambulance relocation and resource allocation under uncertainty.

University of Waterloo

Graduate Teaching Assistant, Faculty of Mathematics

Waterloo, Canada

Sep 2025 – Present

- Supported MATH 128, MATH 106, MATH 135, and MATH 115 through tutorials, office hours, grading, and feedback on calculus, linear algebra, and proofs.

Amirkabir University of Technology

Teaching Assistant

Tehran, Iran

Sep 2023 – Jun 2025

- Supported Probability Theory, Optimization, and Systems Analysis; designed mathematical-modeling assignments.

TAV Institute (Tavenergyco)

Factory Layout Design Intern

Tehran, Iran

2024

- Conducted factory-layout analysis, process-flow modeling, and material-handling capacity analysis.

SELECTED PROJECTS

Dynamic Ambulance Dispatch and Relocation

Python, Discrete-Event Simulation, Optimization

- Developed an event-driven EMS simulator representing stochastic call arrivals, dispatch, service progression, ambulance availability, and preemptible relocation.
- Implemented an initial interpretable MOVE-versus-STAY framework that evaluates relocation trajectories, spatial demand, local coverage, and interactions among available ambulances.
- Exploring stochastic optimization, structured combinatorial search, and learning-guided action evaluation as methodological extensions.

CrashVision-DL: Traffic Severity Prediction

PyTorch, DeepInsight, Imbalanced Learning

- Built a tabular-to-image pipeline on 7.7M records with DeepInsight feature mapping and a Soft F-Beta objective for severe-crash recall.

Academic Hub Automation & Scientific Content Pipeline

Docker, AsyncIO, Microsoft 365

- Built Linux–Microsoft 365 research automation and an asynchronous pipeline converting scientific PDFs, including math content, into audio notes.

TECHNICAL SKILLS

Languages Python, C/C++, Bash

Methods Numerical/combinatorial optimization, stochastic modeling, discrete-event simulation, policy evaluation

ML / AI PyTorch, scikit-learn, XGBoost, EfficientNet, DeepInsight

Systems Linux, Docker, Bash automation, AsyncIO, networking protocols

HONORS AND AWARDS

- i-Comp-Data Fellowship**, University of Waterloo; ranked in the top 1% among approximately 150,000 candidates in the National University Entrance Examination; Merit Scholarship, Amirkabir University of Technology.